Job Market Reality Check — README

Overview

This project demonstrates an end-to-end Data Analytics workflow using Python, SQL, Excel, and Power BI to analyze real-world Data Analyst job postings and generate insights on in-demand skills, salary drivers, and regional hiring trends.

Dataset

Job title, company name, location (city/state), salary estimate, company rating, industry, sector, company size, type of ownership, and full job description text (used to extract required skills).

Tools

Python, Pandas, NumPy, Regex, PostgreSQL/MySQL, Excel, Power BI, Jupyter Notebook/VS Code, Git & GitHub.

Steps

1. Load dataset.

2. Clean data.

3. Extract skills from job descriptions.

4. Run SQL queries.

5. Build Excel pivot reports.

6. Build Power BI dashboard.

7. Prepare report and README.

Dashboard

KPIs, Skill Demand Overview, Salary by Skill, Top Cities, Skill Demand by City, Salary by Seniority Level, and Interactive Filters (Skill, Rating, State, City).

Results

Identified the most in-demand skills, skill combinations linked to higher salaries, top hiring cities, regional differences in skill demand, and the salary gap between junior and senior roles — along with career recommendations based on the findings.

How to Run

Install required libraries (pandas, numpy, re, sqlalchemy), run the Python cleaning/skill-extraction script, load the cleaned data into SQL and execute the analysis queries, open the Excel workbook to review pivot tables, open the Power BI (.pbix) file to explore the dashboard, and review the accompanying report.

Author

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